OTAGIL 10 mg Tablet

OTAGIL 20 mg Tablet

HMG-CoA REDUCTASE INHIBITOR

Each film coated tablet contains:
Rosuvastatin calcium eq. to Rosuvastatin 10 mg
Rosuvastatin calcium eq. to Rosuvastatin 20 mg

✧ PRODUCT DESCRIPTION:

10 mg:
Pink, round, biconvex, film coated tablet with engraved RSV and 10 on one side and plain on the other [7.00 mm (± 0.50 mm) diameter]

20 mg:
Pink, round, biconvex, film coated tablet with engraved RSV and 20 on one side and plain on the other [9.00 mm (± 0.50 mm) diameter]

✧ MECHANISM OF ACTION:

Pharmacology

Rosuvastatin is a selective and competitive inhibitor of HMG-CoA reductase, the rate-limiting enzyme that converts 3-hydroxy-3-methylglutaryl coenzyme A to Mevalonate, a precursor for cholesterol. Triglycerides (TG) and cholesterol in the liver are incorporated, with apolipoprotein B (ApoB), into very low density lipoprotein (VLDL) and released into the plasma for delivery to peripheral tissues. VLDL particles are TG-rich. Cholesterol-rich low density lipoprotein (LDL) is formed from VLDL and is cleared primarily through the high affinity LDL receptor in the liver.

Rosuvastatin produces its lipid-modifying effects in two ways: it increases the number of hepatic LDL receptors on the cell-surface, enhancing uptake and catabolism of LDL and it inhibits the hepatic synthesis of VLDL, thereby reducing with total number of VLDL and LDL particles.

High density lipoprotein (HDL), which contains ApoA-I is involved, amongst other things, in transport of cholesterol from tissues back to the liver (reverse cholesterol transport).The involvement of LDL-C in atherogenesis has been well documented. Epidemiological studies have established that high LDL-C, TG, low HDL-C and ApoA-1 have been linked to a higher risk of cardiovascular disease. Intervention studies have shown the benefits on mortality and CV event rates of lowering LDL-C and TG or raising HDL-C. More recent data has linked the beneficial effects of HMG-CoA reductase inhibitors to lowering of non-HDL (i.e. all circulating cholesterol not in HDL) and ApoB or reducing the ApoB/ApoA-I ratio.

Pharmacokinetics

Absorption

In man, peak plasma concentration of Rosuvastatin were reached 3 to 5 hours following oral dosing. Both peak concentration (C_{max}) and area under the plasma concentration-time curve (AUC) increased in approximate proportion to Rosuvastatin dose. The absolute bioavailability of Rosuvastatin is approximately 20%. Administration of Rosuvastatin with food decreased the rate of drug absorption by 20% as assessed by C_{max}, but there was no effect on the extent of absorption as assessed by AUC.

Plasma concentrations of Rosuvastatin do not differ following evening or morning drug administration. Significant LDL-C reductions are seen when Rosuvastatin is given with or without food, and regardless of the time of day of drug administration.

Distribution

Mean volume of distribution at steady-state of Rosuvastatin is approximately 134 Liters. Rosuvastatin is 88% bound to plasma proteins, mostly albumin. This binding is reversible and independent of plasma concentrations.

Metabolism

Rosuvastatin is not extensively metabolized; approximately 10% of a radio-labelled dose is recovered as metabolite. The major metabolite is N-desmethyl rosuvastatin, which is formed principally by cytochrome P450 2C9, and *in vitro* demonstrated that N-desmethyl Rosuvastatin has approximately one-sixth to one-half the HMG-CoA reductase inhibitory activity of Rosuvastatin. Overall, greater than 90% of active plasma HMG-CoA reductase inhibitory activity is accounted for by Rosuvastatin.

Excretion

Following oral administration, Rosuvastatin and its metabolite are primarily excreted in the feces (90%). The elimination half-life ($t_{1/2}$) of Rosuvastatin is approximately 19 hours.

After an intravenous dose, approximately 28% of total body clearance was via the renal route and 72% by the hepatic route.

Special populations

Age and sex

There was no clinically relevant effect of age or sex on the pharmacokinetics of Rosuvastatin in adults.

Pediatric use

The safety and effectiveness of Rosuvastatin in patients 10 to 17 years of age with heterozygous familial hypercholesterolemia were evaluated. Patients treated with 10 mg and 20 mg daily Rosuvastatin had an adverse experience profile generally similar to that of patients treated with placebo. Although not all adverse reactions identified in the adult population have been observed in children and adolescent patients, the same warnings and precautions for adults should be considered for children and adolescents. There was no detectable effect of Rosuvastatin on growth, weight, BMI (body mass index), or sexual maturation in pediatric patients (10 to 17 years of age). Adolescent females should be counseled on appropriate contraceptive methods while on Rosuvastatin therapy.

Pediatric patients (10 to 17 years of age) with heterozygous FH received single and multiple oral doses of Rosuvastatin. Both C_{max} and AUC of Rosuvastatin were similar to values observed in adult subjects administered the same doses.

Genetic polymorphisms

Disposition of HMG-CoA reductase inhibitors, including Rosuvastatin, involves OATP1B1 and BCRP transporter proteins. In patients with SLCO1B1 (OATP1B1) and/or ABCG2 (BCRP) genetic polymorphisms there is a risk of increased Rosuvastatin exposure. Individual polymorphisms of SLCO1B1 c.521CC and ABCG2 c.421AA are associated with an approximate 1.6 fold higher Rosuvastatin exposure (AUC) compared to the SLCO1B1 c.521TT or ABCG2 c.421CC genotypes. This specific genotyping is not established, but for patients who are known to have these types of polymorphisms, a lower daily dose of Rosuvastatin is recommended.

Race

An approximate 2-fold elevation in median AUC and C_{max} in Asian subjects compared with Caucasians. A population pharmacokinetic analysis revealed no clinically relevant differences in pharmacokinetics between Caucasian, Hispanic and Black or Afro-Caribbean groups.

Patients with renal impairment

Mild to moderate renal impairment ($CL_{CR} \geq 30$ mL/min/1.73 m^2) had no influence on plasma concentrations of Rosuvastatin. However, plasma concentrations of Rosuvastatin increased to a clinically significant extent (about 3-fold) in patients with severe renal impairment ($CL_{CR} < 30$ mL/min/1.73 m^2) not receiving hemodialysis compared with healthy subjects ($CL_{CR} > 80$ mL/min/1.73 m^2).

✧ INDICATION:

Otagil (Rosuvastatin) is indicated as an adjunct to diet, at least equivalent to the Adult Treatment Panel III (ATP III TLC diet), for the reduction of elevated total cholesterol, LDL-cholesterol, ApoB, the total cholesterol: HDL-cholesterol ratio and triglycerides and for increasing HDL-C, in hyperlipidemic and dyslipidemic conditions, when response to diet and exercise alone has been inadequate including:

- Prevention of Cardiovascular Events. In adult patients with an increased risk of atherosclerotic cardiovascular disease based on the presence of cardiovascular disease risk markers such as an elevated hsCRP level, age, hypertension, low HDL-C, smoking or a family history of premature coronary heart disease, Rosuvastatin is indicated to reduce total mortality and the risk of major cardiovascular events (cardiovascular death, stroke, MI, unstable angina, or arterial revascularization).
- Indicated as an adjunct to diet for the treatment of patients with primary dysbetalipoproteinemia (Type III Hyperlipoproteinemia).
- Primary hypercholesterolemia (Type IIa including heterozygous familial hypercholesterolemia and severe non-familial hypercholesterolemia).
- Combined (mixed) dyslipidemia (Type IIb)
- Homozygous familial hypercholesterolemia where Rosuvastatin is used either alone or as an adjunct to diet and other lipid lowering treatment such as apheresis.
- Indicated as adjunctive therapy to diet to slow the progression of atherosclerosis in adult patients as part of treatment strategy to lower Total-C and LDL-C to target levels.
- Pediatric patients 10 to 17 years of age with Heterozygous Familial Hypercholesterolemia (HeFH): Adjunct to diet to reduce Total-C, LDL-C and ApoB levels in adolescent boys and girls, who are at least one year post-menarche, 10-17 years of age with heterozygous familial hypercholesterolemia if after an adequate trial of diet therapy the following findings are present: LDL-C > 190 mg/dL or 160 mg/dL and there is a positive family history of premature cardiovascular disease (CVD) or two or more other CVD risk factors.

✧ DOSAGE AND ADMINISTRATION:

Oral

Patients should be placed on a standard cholesterol-lowering diet (at least equivalent to the Adult Treatment Panel III (ATP III TLC diet)) before receiving Otagil (Rosuvastatin calcium), and should continue on this diet during treatment with Otagil. If appropriate, a program of weight control and physical exercise should be implemented.

Prior to initiating therapy with Otagil, secondary causes for elevations in plasma lipid levels should be excluded. A lipid profile should also be performed. After initiation or upon titration of Otagil, lipid levels should be analyzed within 2-4 weeks and the dosage adjusted accordingly.

The usual recommended starting dose of Otagil is 10 mg once daily. However, initiation of therapy with 5 mg once daily should be considered for special patient populations or patients requiring less aggressive LDL-C reductions. The choice of starting dose should take into account the individual patients' cholesterol level and future cardiovascular risk as well as the potential risk for adverse reactions. Otagil may be taken in the morning or evening, with or without food. The majority of patients are controlled at the 10 mg dose. However, if necessary, dose adjustments to the next dose level can be made after 4-week intervals. The maximum response is usually achieved within 2-4 weeks and is maintained during chronic therapy. Increasing the dose to 40 mg should be reserved for patients with severe hypercholesterolemia at high cardiovascular risk (in particular those with familial hypercholesterolemia), who do not achieve their treatment goal on 20 mg and should only be initiated under specialist supervision (see *Warning and Precaution*). The physician who elects to use Otagil at a dose higher than 20 mg should periodically re-evaluate the long term risk/benefit of Otagil for the individual patient. Otagil should be prescribed with caution in patients with pre-disposing factors for myopathy / rhabdomyolysis (see *Warning and Precaution*).

The dosage of Otagil should be individualized according to baseline LDL-C, total-C/HDL-C ratio and/or TG levels, the recommended target lipid values (see *Recommendations for the Management and Treatment of Dyslipidemia [Canada]* summarized below in Table 1) and/or the Third Report of the U.S. National Cholesterol Education Program [NCEP Adult Treatment Panel III]) and the patient response.

The majority (80%) of patients treated with Rosuvastatin 10 mg achieved their NCEP ATP III treatment target for LDL-C levels; fewer subjects (68%) achieved target on the 5 mg dose. The difference between Rosuvastatin 5 mg and 10 mg was greatest for high risk subjects (40% versus 61%, respectively), i.e. for subjects who have a lower LDL-C target.

Lipid levels should be monitored periodically and, if necessary, the dose of Otagil adjusted based on target lipid levels recommended by guidelines.

Table 1: Canadian Recommendation for Target Lipid Values Based on Level of Risk

Level of Risk (definition)	Target values LDL-C (mmol/L)	Total-C/HDL-C ratio	TG (mmol/L)
Very high* (10-year risk of CAD > 30%, or history of cardiovascular disease or diabetes)	< 2.5	< 4.0	< 2.0
High* (10-year risk CAD 20% - 30%)	< 3.0	< 5.0	< 2.0
Moderate** (10-year risk of CAD 10% - 20%)	< 4.0	< 6.0	< 2.0
Low*** (10-year risk of CAD <10%)	< 5.0	< 7.0	< 3.0

* Start medication and lifestyle changes concomitantly if values are above target values
** Start medication if target values are not achieved after 3 months of lifestyle modification
*** Start medication if target values are not achieved after 6 months of lifestyle modification

The following reductions in total cholesterol, LDL-C, TG, Total-C/HDL and increases in HDL-C have been observed in a dose-response study, and may serve as a guide to treatment of patients with mild to moderate hypercholesterolemia:

Table 2: Dose-Response in Patients with Mild to Moderate Hypercholesterolemia (Mean Percent change from Baseline)

Otagil dose (mg/day)	N	Total-C	LDL-C	TG	HDL-C	Total-C/HDL-C	Apo B
Placebo	13	- 5	- 7	- 3	3	- 8	- 3
5	17	- 33	- 45	- 35	13	- 41	- 38
10	17	- 36	- 52	- 10	14	- 43	- 42
20	17	- 40	- 55	- 23	8	- 44	- 46
40	18	- 46	- 63	- 28	10	- 51	- 54

Dosage in patients with renal insufficiency

The usual dose range applies in patients with mild to moderate renal impairment.

For patients with severe renal impairment ($CL_{cr} < 30$ mL/min/1.73 m^2) not on hemodialysis, dosing of Otagil should be started at 5 mg once daily and not exceed 10 mg once daily (see *Pharmacokinetics*).

Dosage in patients with hepatic insufficiency

There was no increase in systemic exposure to Rosuvastatin in subjects with Child-Pugh scores of 7 or below. However, increased systemic exposure has been observed in subjects with Child-Pugh scores of 8 and 9. In these patients an assessment of renal function should be considered. There is no experience in subjects with Child-Pugh scores above 9. Otagil is contraindicated in patients with active liver disease.

Use in the elderly

Of the 10,275 patients in clinical studies with Rosuvastatin, 3,159 (31%) were 65 years and older, and 698 (6.8%) were 75 years and older. The overall frequency of adverse events and types of adverse events were similar in patients above and below 65 years of age. The efficacy of Rosuvastatin in the geriatric population (≥ 65 years of age) was comparable to the efficacy observed in the non-elderly.

Pediatric patients (10 to 17 years of age)

In pediatric patients (10 to 17 years of age) with heterozygous familial hypercholesterolemia the usual dose range of Otagil is 5-20 mg/day; the maximum recommended dose is 20 mg/day (doses greater than 20 mg have not been studied in this patient population). Doses should be individualized according to the recommended goal of therapy (see *Indication*). Adjustments should be made at intervals of 4 weeks or more.

Use in children below 10 years

The safety and effectiveness in children have not been established. In children and adolescents with homozygous familial hypercholesterolemia experience is limited to eight patients (aged 8 years and above).

Dosage on Asian patients

Initiation of Otagil therapy with 5 mg once daily should be considered for Asian patients. The potential for increased systemic exposures relative to Caucasians is relevant when considering escalation of dose in cases where hypercholesterolemia is not adequately controlled at doses of 5, 10 or 20 mg once daily (see *Warning and Precaution* and *Pharmacokinetics*).

Genetic polymorphisms

Specific types of genetic polymorphisms are known that can lead to increased Rosuvastatin exposure (see *Pharmacokinetics*). For patients who are known to have such specific types of polymorphisms, a lower daily dose of Otagil is recommended.

Dosage in patients with pre-disposing factors to myopathy

The recommended start dose is 5 mg in patients with predisposing factors to myopathy (see *Warning and Precaution*).

Concomitant therapy

Rosuvastatin is a substrate of various transporter proteins (e.g. OATP1B1and BCRP). The risk of myopathy (including rhabdomyolysis) is increased when Otagil is administered concomitantly with certain medicinal products that may increase the plasma concentration of Rosuvastatin due to interactions with these transporter proteins (e.g. Ciclosporin and certain protease inhibitors including combinations of Ritonavir with Atazanavir, Lopinavir, and/or Tipranavir; (see *Warning and Precaution* and *Drug interaction*). Whenever possible, alternative medications should be considered, and, if necessary, consider temporarily discontinuing Otagil therapy. In situations where co-administration of these medicinal products with Otagil is unavoidable, the benefit and the risk of concurrent treatment and Otagil dosing adjustments should be carefully considered (see *Drug interaction*). If concomitant use of Otagil with Ciclosporin cannot be avoided, the dose of Otagil should not exceed 5 mg once daily (see *Warning and Precaution* and *Drug interaction*).

✧ WARNING AND PRECAUTION:

Renal Effects

Proteinuria, detected by dipstick testing and mostly tubular in origin, has been observed in patients treated with higher doses of Rosuvastatin, in particular 40 mg, where it was transient or intermittent in most cases. Proteinuria has not been shown to be predictive of acute or progressive renal disease. An assessment of renal function should be considered during routine follow-up of patients treated with a dose of 40 mg.

Renal impairment

Rosuvastatin exposure is not influence by mild to moderate renal impairment ($CL_{CR} \geq 30$ mL/min/1.73 m^2). Exposure to Rosuvastatin is increased to a clinically significant extent in patients with severe renal impairment ($CL_{CR} < 30$ mL/min/1.73 m^2) who are not receiving hemodialysis and dose adjustment is required (see *Dosage and administration* and *Pharmacokinetics*).

Skeletal Muscle Effects

Gemfibrozil increases the risk of myopathy when given concomitantly with some HMG-CoA reductase inhibitors. Therefore, the combination of Rosuvastatin and Gemfibrozil is not recommended. The benefit of further alterations in lipid levels by the combined use of Rosuvastatin with fibrates or Niacin should be carefully weighed against the potential risks of such combinations.

Effects on skeletal muscle e.g. myalgia, myopathy and, rarely, rhabdomyolysis have been reported in Rosuvastatin-treated patients with all doses and in particular with doses > 20 mg. Very rare cases of rhabdomyolysis have been reported with the use of Ezetimibe in combination with HMG-CoA reductase inhibitors. A pharmacodynamic interaction cannot be excluded (see *Drug interaction*) and caution should be exercised with their combined use. As with other HMG-CoA reductase inhibitors, the reporting rate for rhabdomyolysis associated with Rosuvastatin in post-marketing use is higher at the 40 mg dose.

Creatine Kinase Measurement

Creatine Kinase (CK) should not be measured following strenuous exercise or in the presence of a plausible alternative cause of CK increase which may confound interpretation of the result. If CK levels are significantly elevated at baseline (> 5 x ULN) a confirmatory test should be carried out within 5-7 days. If the repeat test confirms a baseline CK > 5 x ULN, treatment should not be started.

Before Treatment

Rosuvastatin, as with other HMG-CoA reductase inhibitors, should be prescribed with caution in patients with pre-disposing factors for myopathy/rhabdomyolysis. Such factors include:

- renal impairment
- hypothyroidism
- personal or family history of hereditary muscular disorders
- previous history of muscular toxicity with another HMG-CoA reductase inhibitor or fibrate
- alcohol abuse
- age > 70 years
- situations where an increase in plasma levels may occur
- concomitant use of fibrates

In such patients the risk of treatment should be considered in relation to possible benefit and clinical monitoring is recommended. If CK levels are significantly elevated at baseline (> 5 x ULN) treatment should not be started.

Whilst on Treatment

Patients should be asked to report inexplicable muscle pain, weakness or cramps immediately, particularly if associated with malaise or fever. CK levels should be measured in these patients. Therapy should be discontinued if CK levels are markedly elevated (> 5 x ULN) or if muscular symptoms are severe and cause daily discomfort (even if CK levels are ≤ 5 x ULN). If symptoms resolve and CK levels return to normal, then consideration should be given to re-introducing Rosuvastatin or an alternative HMG-CoA reductase inhibitor at the lowest dose with close monitoring. Routine monitoring of CK levels in asymptomatic patients is not warranted.

There have been very rare reports of an immune-mediated necrotizing myopathy (IMNM) during or after treatment with statins, including Rosuvastatin. IMNM is clinically characterized by:

- persistent proximal muscle weakness and elevated serum creatinine kinase, which persist despite discontinuation of statin treatment
- muscle biopsy showing necrotizing myopathy without significant inflammation
- improvement with immunosuppressive agents

There was no evidence of increased skeletal muscle effects in the small number of patients dosed with Rosuvastatin and concomitant therapy. However, an increase in the incidence of myositis and myopathy has been seen in patients receiving other HMG-CoA reductase inhibitors together with Fibric acid derivatives including Gemfibrozil, Ciclosporin, Nicotinic acid, azole antifungals, protease inhibitors and macrolide antibiotics. Gemfibrozil increases the risk of myopathy when given concomitantly with some HMG-CoA reductase inhibitors.Therefore, the combination of Rosuvastatin and Gemfibrozil is not recommended. The benefit of further alterations in lipid levels by the combined use of Rosuvastatin with fibrates or Niacin should be carefully weighed against the potential risks of such combinations.

Front
20 cm.
25 cm.
OTAGIL 10,20_Insert_Front.Back_IOMY 0032
BY ..Nuy 19/05/2025
Print: Fit To Page

PM Specification

Product	Code No.	Dimension	Packaging Type	Thickness
OTAGIL 10,20	IOMY 0032	W. 20 X L. 25 cm.	Wood Free Paper (กระดาษปอนด์)	60 g (0.08 mm)

Designed by: (PDM, ASPD, PDC-M, PDC-A, PDO-M, PDO-A)
Checked by: (PDM/ASPD)
Approved by: (MD (Only Pattern and Colors))
Approved by:
PLC: (Date)
LRA: (Date)
IRA: (Date)
QCC-PM: (Date)
CUSTOMER/SALES DEPT.: (Date)

Rosuvastatin should not be used in any patient with an acute, serious condition suggestive of myopathy or predisposing to the development of renal failure secondary to rhabdomyolysis (e.g. sepsis, hypotension, major surgery, trauma, severe metabolite, endocrine and electrolyte disorders; or uncontrolled seizures).

Liver Effects

As with other HMG-CoA reductase inhibitors, Rosuvastatin should be used with caution in patients who consume excessive quantities of alcohol and/or have a history of liver disease.

It is recommended that liver function tests be carried out prior to, and 3 months following, the initiation of treatment. Rosuvastatin should be discontinued or the dose reduced if the level of serum transaminases is greater than 3 times the upper limit of normal. The reporting rate for serious hepatic events (consisting mainly of increased hepatic transaminases) in post-marketing use is higher at the 40 mg dose.

In patients with secondary hypercholesterolemia caused by hypothyroidism or nephrotic syndrome, the underlying disease should be treated prior to initiating therapy with Rosuvastatin.

Race

Pharmacokinetic show an increase in exposure in Asian subjects compared with Caucasians.

Protease Inhibitors

Increased systemic exposure to Rosuvastatin has been observed in subjects receiving Rosuvastatin concomitantly with various protease inhibitors in combination with Ritonavir. Consideration should be given both to the benefit of lipid lowering by use of Rosuvastatin in HIV patients receiving protease inhibitors and the potential for increased Rosuvastatin plasma concentrations when initiating and up titrating Rosuvastatin doses in patients treated with protease inhibitors. The concomitant use with certain protease inhibitors is not recommended unless the dose of Rosuvastatin is adjusted.

Lactose Intolerance

Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Interstitial Lung Disease

Exceptional cases of interstitial lung disease have been reported with some statins, especially with long-term therapy.

Presenting features can include dyspnea, non-productive cough and deterioration in general health (fatigue, weight loss and fever). If it is suspected a patient has developed interstitial lung disease, statin therapy should be discontinued.

Diabetes Mellitus

Some evidence suggests that statins as a class raise blood glucose and in some patients, at high risk of future diabetes, may produce a level of hyperglycemia where formal diabetes care is appropriate. This risk, however, is outweighed by the reduction in vascular risk with statins and therefore should not be a reason for stopping statin treatment. Patients at risk (fasting glucose 5.6 to 6.9 mmol/L, BMI > 30 Kg/m², raised triglycerides, hypertension) should be monitored both clinically and biochemically according to national guidelines.

Pediatric patients (10 to 17 years of age)

The evaluation of linear growth (height), weight, BMI (body mass index), and secondary characteristics of sexual maturation by Tanner staging in pediatric patients taking Rosuvastatin is limited to a one year period.

Myasthenia Gravis/ Ocular Myasthenia

In few cases, statins have been reported to induce de novo or aggravate pre-existing myasthenia gravis or ocular myasthenia. Otagil should be discontinued in case of aggravation of symptoms. Recurrences when the same or a different statin was (re-) administered have been reported.

✧ PREGNANCY AND LACTATION:

The safety of Rosuvastatin during pregnancy and whilst breastfeeding has not been established. Women of child-bearing potential should use appropriate contraceptive measures (see *Contraindication)*.

✧ SIDE EFFECT:

Tabulated list of adverse reactions:

System Organ Class	Common	Uncommon	Rare	Very rare	Not known
Blood and lymphatic system disorders			Thrombocytopenia		
Immune system disorders			Hypersensitivity reactions including angioedema		
Endocrine disorders	Diabetes mellitus[1]				
Psychiatric disorders					Depression
Nervous system disorders	Headache Dizziness			Polyneuropathy Memory loss	Sleep disturbances (including insomnia and nightmares) Peripheral neuropathy Myasthenia gravis
Eye disorders					Ocular myasthenia
Respiratory, thoracic and mediastinal disorders					Cough Dyspnea
Gastrointestinal disorders	Constipation Nausea Abdominal pain		Pancreatitis		Diarrhea
Hepatobiliary disorders			Increased hepatic transaminases	Jaundice Hepatitis	
Skin and subcutaneous tissue disorders		Pruritus Rash Urticaria			Stevens-Johnson syndrome
Musculoskeletal and connective tissue disorders	Myalgia		Myopathy (including myositis) Rhabdomyolysis	Arthralgia	Immune-mediated necrotizing myopathy
Renal and urinary disorders				Hematuria	
Reproductive system and breast disorders				Gynecomastia	
General disorders and administration site conditions	Athenia				Edema

[1]Frequency will depend on the presence or absence of risk factors (fasting blood glucose ≥ 5.6 mmol/L, BMI > 30 Kg/m², raised triglycerides, history of hypertension).

As with other HMG-CoA reductase inhibitors, the incidence of adverse drug reactions tends to be dose dependent.

Renal effects

Proteinuria, detected by dipstick testing and mostly tubular in origin, has been observed in patients treated with Rosuvastatin. Shifts in urine protein from none or trace to ++ or more were seen in < 1% of patients at some time during treatment with 10 and 20 mg, and in approximately 3% of patients treated with 40 mg. A minor increase in shift from none or trace to + was observed with the 20 mg dose. In most cases, proteinuria decreases or disappears spontaneously on continued therapy.

Hematuria has been observed in patients treated with Rosuvastatin and data show that the occurrence is low.

Skeletal muscle effects

Effects on skeletal muscle e.g. myalgia, myopathy (including myositis) and, rarely, rhabdomyolysis with and without acute renal failure have been reported in Rosuvastatin-treated patients with all doses and in particular with doses > 20 mg.

A dose-related increase in CK levels has been observed in patients taking Rosuvastatin; the majority of cases were mild, asymptomatic and transient. If CK levels are elevated (> 5 x ULN), treatment should be discontinued.

Liver effects

As with other HMG-CoA reductase inhibitors, a dose-related increase in transaminases has been observed in a small number of patients taking Rosuvastatin; the majority of cases were mild, asymptomatic and transient.

The following adverse events have been reported with some statins:
- Sexual dysfunction
- Exceptional cases of interstitial lung disease, especially with long term therapy
- Tendon disorders, sometimes complicated by rupture.

The reporting rates for rhabdomyolysis, serious renal events and serious hepatic events (consisting mainly of increased hepatic transaminases) is higher at the 40 mg dose.

Pediatric patients 10 to 17 years of age

Elevations in serum creatine phosphokinase (CK) > 10 x ULN were observed more frequently in Rosuvastatin compared with placebo-treated children.

There have been rare post-marketing reports of cognitive impairment (e.g. memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use. These cognitive issues have been reported for all statins. The reports are generally non-serious and reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median 3 weeks).

Increases in HbA1c and fasting blood glucose have been reported with statins. The risk of hyperglycemia, however, is outweighed by the reduction in vascular risk with statins.

✧ EFFECTS ON ABILITY TO DRIVE AND USE MACHINES:

Not applicable.

✧ CONTRAINDICATION:

Otagil is contraindicated:
- in patients with hypersensitivity to Rosuvastatin or to any component of this product.
- in patients with active liver disease including unexplained, persistent elevations of serum transaminases and any serum transaminase elevation exceeding 3 times the upper limit of normal (ULN).
- during pregnancy, while breastfeeding and in women of child-bearing potential not using appropriate contraceptive measures.
- in patients with myopathy.

Otagil is contraindicated in patients receiving concomitant Cyclosporin.

✧ DRUG INTERACTION:

Effect of co-administered medicinal products on Rosuvastatin

Transporter protein inhibitors

Rosuvastatin is a substrate for certain transporter proteins including the hepatic uptake transporter OATP1B1 and efflux transporter BCRP.

Concomitant administration of Otagil with medicinal products that are inhibitors of these transporter proteins may result in increased Rosuvastatin plasma concentrations and an increased risk of myopathy (see *Dosage and administration, Warning and Precaution* and *Drug interaction Table 3*).

Ciclosporin

Ciclosporin increased Rosuvastatin exposure and may result in increased risk of myopathy (see Table 3). Therefore, in patients taking Ciclosporin, the dose of Otagil should not exceed 5 mg once daily (*see Dosage and administration* and *Warning* and *Precaution*).

Protease inhibitors

Although the exact mechanism of interaction is unknown, concomitant protease inhibitor use may strongly increase Rosuvastatin exposure (see Table 3). For instance, in a pharmacokinetic study, co-administration of 10 mg Rosuvastatin and a combination product of two protease inhibitors (300 mg Atazanavir/ 100 mg Ritonavir) in healthy volunteers was associated with an approximately three-fold and seven-fold increase in Rosuvastatin AUC and C_{max} respectively. The concomitant use of Otagil and some protease inhibitor combinations may be considered after careful consideration of Otagil dose adjustments based on the expected increase in Rosuvastatin exposure (see *Dosage and administration, Warning and Precaution* and *Drug interaction Table 3*).

Gemfibrozil and other lipid-lowering products

Concomitant use of Otagil and Gemfibrozil resulted in a 2-fold increase in Rosuvastatin C_{max} and AUC (see *Warning and Precaution*). Based on data from specific interaction studies no pharmacokinetic relevant interaction with Fenofibrate is expected, however a pharmacodynamic interaction may occur. Gemfibrozil, Fenofibrate, other fibrates and lipid lowering doses (> or equal to 1g/day) of niacin (nicotinic acid) increase the risk of myopathy when given concomitantly with HMG-CoA reductase inhibitors, probably because they can produce myopathy when given alone. These patients should also start with the 5 mg dose.

Ezetimibe

Concomitant use of 10 mg Otagil and 10 mg Ezetimibe resulted in a 1.2 fold increase in AUC of Rosuvastatin in hypercholesterolemic subjects (Table 3). A pharmacodynamic interaction, in terms of adverse effects, between Rosuvastatin and Ezetimibe cannot be ruled out (see *Warning and Precaution*).

Antacid

The simultaneous dosing of Otagil with an antacid suspension containing Aluminium and Magnesium hydroxide resulted in a decrease in Rosuvastatin plasma concentration of approximately 50%. This effect was mitigated when the antacid was dosed 2 hours after Otagil. The clinical relevance of this interaction has not been studied.

Fusidic acid

Interaction studies with Rosuvastatin and Fusidic acid have not been conducted. As with other statins, muscle related events, including rhabdomyolysis, have been reported in post-marketing experience with Rosuvastatin and Fusidic acid given concurrently. Patients should be closely monitored and temporary suspension of Rosuvastatin treatment may be appropriate.

Erythromycin

Concomitant use of Otagil and Erythromycin resulted in a 20% decrease in $AUC_{(0-t)}$ and a 30% decrease in C_{max} of Rosuvastatin. This interaction may be caused by the increase in gut motility caused by Erythromycin.

Cytochrome P450 enzymes

Rosuvastatin is neither an inhibitor nor an inducer of cytochrome P450 isoenzymes. In addition, Rosuvastatin is a poor substrate for these isoenzymes. Therefore, drug interactions resulting from cytochrome P450-mediated metabolism are not expected. No clinically relevant interactions have been observed between Rosuvastatin and either Fluconazole (an inhibitor of CYP2C9 and CYP3A4) or Ketoconazole (an inhibitor of CYP2A6 and CYP3A4).

Interactions requiring Rosuvastatin dose adjustments (see also Table 3)

When it is necessary to co-administer Otagil with other medicinal products known to increase exposure to Rosuvastatin, doses of Otagil should be adjusted. It is recommended that prescribers consult the relevant product information when considering administration of such products together with Otagil.

If medicinal product is observed to increase Rosuvastatin AUC approximately 2-fold or higher, the starting dose of Otagil should not exceed 5 mg once daily. The maximum daily dose of Otagil should be adjusted so that the expected Rosuvastatin exposure would not likely exceed that of a 40 mg daily dose of Otagil taken without interacting medicinal products, for example a 5 mg dose of Otagil with Ciclosporin (7.1-fold increase in exposure), a 10 mg dose of Otagil with Ritonavir/ Atazanavir combination (3.1-fold increase) and a 20 mg dose of Otagil with Gemfibrozil (1.9-fold increase).

If medicinal product is observed to increase Rosuvastatin AUC less than 2-fold, the starting dose need not be decreased but caution should be taken if increasing the Otagil dose above 20 mg.

Protease Inhibitors

Co-administration of Rosuvastatin with certain protease inhibitors or combination of protease inhibitors may increase the Rosuvastatin exposure, (AUC) up to 7-fold (see Table 1). Dose adjustment are needed depending on the level of effect on Rosuvastatin exposure (see *Dosage and administration* and *Warning and Precaution*).

Table 3: Effect of co-administered medicinal products on Rosuvastatin exposure (AUC; in order of decreasing magnitude) from published clinical trials

Interacting drug dose regimen	Rosuvastatin dose regimen	Change in Rosuvastatin AUC*
2-fold or greater than 2-fold increase in AUC of Rosuvastatin		
Sofosbuvir/Velpatasvir/Voxilaprevir (400 mg-100 mg-100 mg) + Voxilaprevir (100 mg) once daily for 15 days	10 mg single dose	7.39 -fold ↑
Ciclosporin 75 mg BID to 200 mg BID, 6 months	10 mg OD, 10 days	7.1-fold ↑
Darolutamide 600 mg BID, 5 days	5 mg, single dose	5.2-fold ↑
Regorafenib 160 mg OD, 14 days	5 mg, single dose	3.8 -fold ↑
Atazanavir 300 mg/Ritonavir 100 mg OD, 8 days	10 mg, single dose	3.1-fold ↑
Simeprevir 150 mg OD, 7 days	10 mg, single dose	2.8-fold ↑
Velpatasvir 100 mg OD	10 mg, single dose	2.69 -fold ↑
Ombitasvir 25 mg/Paritaprevir 150 mg/ Ritonavir 100 mg/Dasabuvir 400 mg BID	5 mg, single dose	2.59 -fold ↑
Grazoprevir 200 mg/Elbasvir 50 mg OD	10 mg, single dose	2.26 -fold ↑
Glecaprevir 400 mg/Pibrentasvir 120 mg OD for 7 days	5 mg once daily	2.2 -fold ↑
Lopinavir 400 mg/Ritonavir 100 mg BID, 17 days	20 mg OD, 7 days	2.1-fold ↑
Clopidogrel 300 mg loading, followed by 75 mg at 24 hours [3]	20 mg, single dose[3]	2-fold ↑
Gemfibrozil 600 mg BID, 7 days	80 mg, single dose	1.9-fold ↑
Less than 2-fold increase in AUC of Rosuvastatin		
Interacting drug dose regimen	**Rosuvastatin dose regimen**	**Change in Rosuvastatin AUC***
Eltrombopag 75 mg OD, 5 days	10 mg, single dose	1.6-fold ↑
Darunavir 600 mg/Ritonavir 100 mg BID, 7 days	10 mg OD, 7 days	1.5-fold ↑
Tipranavir 500 mg/Ritonavir 200 mg BID, 11 days	10 mg, single dose	1.4-fold ↑
Dronedarone 400 mg BID	Not available	1.4-fold ↑
Itraconazole 200 mg OD, 5 days	10 mg, single dose	**1.4-fold ↑
Ezetimibe 10 mg OD, 14 days	10 mg OD, 14 days	**1.2-fold ↑
Decrease in AUC of Rosuvastatin		
Interacting drug dose regimen	**Rosuvastatin dose regimen**	**Change in Rosuvastatin AUC***
Erythromycin 500 mg QID, 7 days	80 mg, single dose	20% ↓
Baicalin 50 mg TID, 14 days	20 mg, single dose	47% ↓

* Data given as x-fold change represent a simple ratio between co-administration and Rosuvastatin alone. Data given as % change represent % difference relative to Rosuvastatin alone. Increase is indicated as "↑", decrease as "↓".

** Several interaction studies have been performed at different Otagil dosages, the table shows the most significant ratio

AUC = area under curve; OD = once daily; BID = twice daily; TID = three times daily; QID = four times daily

The following medicinal product/combinations did not have a clinically significant effect on the AUC ratio of Rosuvastatin at co-administration:
- Aleglitazar 0.3 mg 7 days dosing;
- Fenofibrate 67 mg 7 days TID dosing;
- Fluconazole 200 mg 11 days OD dosing;
- Fosamprenavir 700 mg/Ritonavir 100 mg 8 days BID dosing;
- Ketoconazole 200 mg 7 days BID dosing;
- Rifampin 450 mg 7 days OD dosing;
- Silymarin 140 mg 5 days TID dosing

Other medications

Concurrent use of fibrates may cause severe myositis and myoglobinuria.

Effect of Rosuvastatin on co-administered medicinal products

Vitamin K antagonists

As with other HMG-CoA reductase inhibitors, the initiation of treatment or dosage up-titration of Otagil in patients treated concomitantly with Vitamin K antagonists (e.g. Warfarin or another coumarin anticoagulant) may result in an increase in International Normalized Ratio (INR). Discontinuation or down-titration of Otagil may result in a decrease in INR. In such situations, appropriate monitoring of INR is desirable.

Oral contraceptive/hormone replacement therapy (HRT)

Concomitant use of Otagil and an oral contraceptive resulted in an increase in Ethinyl estradiol and Norgestrel AUC of 26% and 34%, respectively. These increased plasma levels should be considered when selecting oral contraceptive doses. There are no pharmacokinetic data available in subjects taking concomitant Otagil and HRT and therefore a similar effect cannot be excluded.

However, the combination has been extensively used in women in clinical trials and was well tolerated.

Other medicinal products

Based on data from specific interaction studies no clinically relevant interaction with Digoxin is expected.

Pediatric population

Interaction studies have only been performed in adults. The extent of interactions in the pediatric population is not known.

✧ OVERDOSE AND TREATMENT:

There is no specific treatment in the event of overdose. In the event of overdose, the patient should be treated symptomatically and supportive measures instituted as required. Liver function and CK levels should be monitored.

Hemodialysis is unlikely to be of benefit.

✧ STORAGE:

Store at temperature of not more than 30ºC.

✧ DOSAGE FORM AND PACKAGING AVAILABLE:

10 mg Tablet, Blister 3x10's
20 mg Tablet, Blister 3x10's

✧ DATE OF REVISION:

May 19, 2025

Manufactured by:
UNISON LABORATORIES CO., LTD.
39 Moo 4, Klong Udomcholjorn, Muang Chachoengsao,
Chachoengsao 24000 Thailand

C-MY/190525-02 (AR)

IOMY 0032

Back 20 cm.
25 cm.

OTAGIL 10,20_Insert_Front.Back_IOMY 0032
BY ..Nuy 19/05/2025

Print: Fit To Page

PM Specification	Product	Code No.	Dimension	Packaging Type	Thickness
	OTAGIL 10,20	IOMY 0032	W. 20 X L. 25 cm.	Wood Free Paper (กระดาษปอนด์)	60 g (0.08 mm)

Designed by: (PDM, ASPD, PDC-M, PDC-A, PDO-M, PDO-A)
Checked by: (PDM/ASPD)
Approved by: (MD (Only Pattern and Colors))

Approved by: PLC: (Date) LRA: (Date) IRA: (Date) QCC-PM: (Date) CUSTOMER/SALES DEPT.: (Date)